

Beyond Text-to-SQL: Reproducing Squirrel

Paper claims verified; benchmark-scale rescoring blocked by the unreleased official dataset.

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· independent reproduction by binzhango

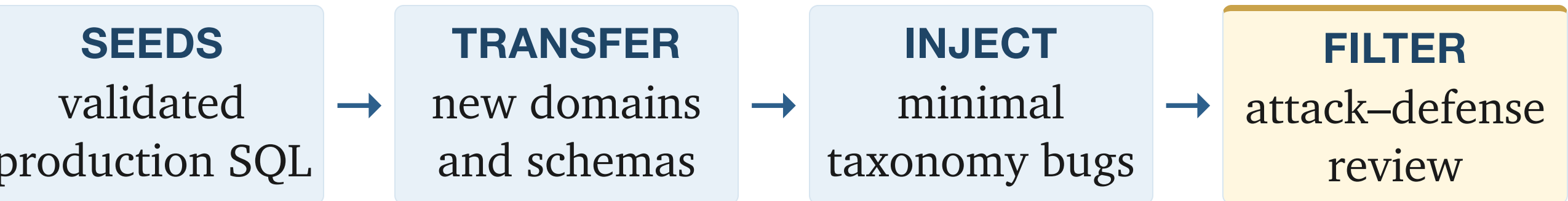
1 What Squirrel claims

Enterprise ETL debugging is not short-query generation. The paper constructs long programs, injects minimal realistic faults, then adversarially filters easy cases.

469	516	1,000+	26
Syntax tasks	Semantic tasks	Seed scripts	Scenarios

Status: all four values occur in the paper; official data were unavailable for row-level recounting.

2 Construction pipeline



- Syntax tasks include explicit analyzer feedback.
- Semantic tasks describe a requirement mismatch.
- Graph Match compares optimized structural representations.

3 Reported enterprise scale

Split	Tokens	Lines	Funcs
Syntax	496.90	163.69	21.62
Semantic	425.93	141.58	17.34

Split	AST depth	AST width
Syntax	8.93	11.69
Semantic	8.75	11.12

The requested “420+ tokens and 17.34–21.62 functions” claim matches Table 1 exactly.

4 Local deterministic proxy

161	1,005	58	12
lines	lexical tokens	sqlglot funcs	toy tasks

10	170	2/2
raw AST depth	raw AST width	smoke tasks passed

Only line count is directly comparable. The paper uses optimized Calcite plans; this proxy uses raw sqlglot ASTs.

Guardrail: the toy proxy tests the workflow, not official benchmark statistics.

5 Headline model result ★ Table 2

36.46%	32.17%	#1
Claude syntax GM	Claude semantic GM	both GM columns

Abstract, Table 2, and Section 5.1 agree on 32.17% semantic GM.

Paper typo: one introduction paragraph says 33.17%; the authoritative table says 32.17%.

6 Open-model checks

Model	Syntax GM	Semantic GM
DeepSeek-V3	30.28%	21.32%
Qwen2.5-Coder-32B	20.26%	23.45%

10/24	15/24	16/24
syntax below 20%	semantic below 20%	either track below 20%

“Most below 20%” holds for semantic GM and either-track failure, but not for syntax GM alone.

7 A10G debugging experiment

Qwen2.5-Coder-7B-Instruct runs greedily on 12 long-SQL tasks: six parser faults and six join-semantics faults.

- Scale:** 12/985 tasks, or 1.2% scale.
- Metric:** strict sqlglot canonical-tree equality proxy.
- Hardware:** one HF A10G-small.

100%	100%	51.42s
syntax GM	semantic GM	mean latency

8 Takeaways ★

- Claims 1–4:** requested paper values are present and internally consistent, except one 33.17 typo.
- Independent audit:** Table 2 narrows how “most below 20%” should be read.
- Experiment:** 12/12 easy toy cases were repaired exactly on an A10G.
- Limit:** official Squirrel examples, predictions, and evaluator remain unavailable.

Bottom line: the numbers verify as reported; public assets do not support full reproduction.